



**Teheran University of Medical
Sciences**

International campus Vice-Dean for Educational Affairs



Teheran University of Medical Sciences

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**Clinical biochemistry
exam- pharmacy**

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Final Exam
International Campus, PharmaD, Program
Clinical biochemistry
February 2024 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Khalili

Date of Exam: Jul 07, 2024

Duration of Exam: 55 Minutes

Number of questions	52
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Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.

1. All hormone receptors are made of:

- a) Lipids
- b) Carbohydrates
- c) Proteins
- d) lipoproteins

2. For steroids hormones, hormone receptor complex can be formed in

- a) Cytoplasm
- b) Nucleus
- c) Blood
- d) Both cytoplasm and blood

3. Second messengers are triggered as a response to:

- a) Steroid hormones
- b) anabolic steroids
- c) peptide hormones
- d) thyroid hormones

4. When stimulating hormone binds to membrane receptor.....

- a) CAMP is broken
- b) G-Protein is activated
- c) steroid hormones enter the cell
- d) adenylate cyclase is deactivated

5. Releasing hormones are produced by the

- a) Hypothalamus
- b) Pituitary
- c) Thyroid
- d) Adrenal cortex

6. Which of the following hormones are released from posterior pituitary gland?

- a) TSH,ACTH
- b) FSH,LH
- c) ADH, Oxytocin
- d) GH, Prolactin

7. POMC can synthesis

- a) FSH
- b) LH
- c) TSH
- d) ACTH

8. Which hormone plays the greatest role in the contractions that occur in the uterus during labor?

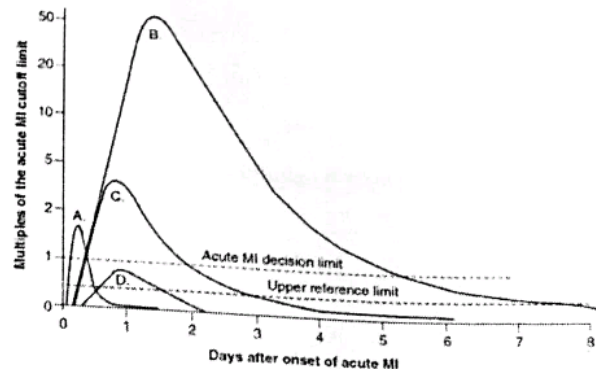
- a) ADH
- b) Estrogen
- c) Oxytocin
- d) Prostaglandin

9. All the following are ways for body water to get out of the body; except?
- Urine
 - Respiration
 - Metabolism
 - Skin
10. Which of the following may not cause hyponatremia?
- Water excess
 - Sodium depletion
 - Water and sodium excess
 - Water depletion
11. A 63-year-old man has the following ABG results. What is his acid-base disorder?
pH 7.22; PaCO₂ 55; HCO₃⁻ 25
- Respiratory acidosis
 - Respiratory alkalosis
 - Metabolic acidosis
 - Metabolic alkalosis
12. Calculate the osmolarity in a patient with Na⁺ = 130 mmol/L; BUN = 34 mmol/L; and glucose = 6mmol/L.
- 170
 - 180
 - 200
 - 300
13. Which of the following is a transport protein for copper?
- Transferrin
 - Metallothionein
 - Hemoglobin
 - Ceruloplasmin
14. The deficiency of all the following elements may cause diabetes-like symptoms, except?
- Chromium
 - Vanadium
 - Potassium
 - Magnesium

15. Which of the following can be used to treat Wilson's disease?
- Copper
 - Zinc
 - Manganese
 - Selenium
16. Which test is most frequently used to measure Glomerular Filtration Rate (GFR)?
- Serum Creatinine
 - Serum Urea
 - Cystatin C
 - Urine Protein
17. Which of the following is correct regarding the sensitivity ranking as the most accurate indicator for GFR assessment?
- Cystatin C- Serum Creatinine- Blood Urea Nitrogen (BUN)-Urine Protein
 - Serum Creatinine- Cystatin C- Blood Urea Nitrogen (BUN)- Urine Protein
 - Urine Protein -Serum Creatinine- Cystatin C- Blood Urea Nitrogen (BUN)- Nitrogen (BUN)- Serum Creatinine- Cystatin C- Urine Protein
 - Nitrogen (BUN)- Serum Creatinine- Cystatin C- Urine Protein
18. Identify the correct statement from the following:
- In healthy individuals, urine contains very little protein.
 - Cystatin C levels are affected by muscle mass or sex.
 - Creatinine production does not correlate with sex.
 - In healthy individuals, urine contains very little glucose.
19. All of the following pairs are sensitive and specific markers for myocardial injury except.?
- cTnT and TnC
 - cTnT and TnI
 - CK-MB and cTnT
 - CK-MB and TnI
20. Which statement regarding B-type natriuretic peptide (BNP) is accurate?
- BNP is released by the kidney during congestive heart failure.
 - BNP may also be elevated in acute, life-threatening pulmonary edema.
 - BNP is the most sensitive marker for heart attacks and strokes.
 - BNP is synthesized as a proactive hormone.

21. In a timeline graph of cardiac markers appearing in blood after symptom onset, which marker is indicated by A, and how is the specificity?

- A) Myoglobin-high specificity
- B) Myoglobin -Low specificity
- C) CK-MB -high specificity
- D) CK-MB-Low specificity



22. All of the laboratory finding align with iron deficiency except:

- A) Low red cell distribution width (RDW)
- B) Low ferritin levels
- C) Low Total Iron Binding Capacity (TIBC)
- D) Low white blood (WBC) count

23. What is the primary cause of macrocytic anemia?

- A) Iron deficiency- Folate deficiency
- B) Vitamin B12 deficiency- Folate deficiency
- C) Folate deficiency- Ferritin deficiency
- D) Ferritin deficiency- Vitamin B12 deficiency

24. Which pair types of anaemia can be triggered by certain drugs?

- A) Immune haemolytic Anaemia-Glucose-6-Phosphate Dehydrogenase (G6PD) deficiency
- B) Iron deficiency- Immune haemolytic anaemia
- C) Folate deficiency -Iron deficiency
- D) Immune haemolytic anaemia- Folate deficiency

25. • _____ is a group of enzymes found mainly in bone, liver, intestines, and placenta.

- A. AST
- B. ALT
- C. GGT
- D. ALP

26. This type of jaundice is due to excessive destruction of RBCs, as seen with sickle cell anemia. In this case, ALT levels do not rise.

- A. Hepatic
- B. Post hepatic
- C. Splenohepatic
- D. Prehepatic

27. Which of the following is not a liver function test?

- A. Albumin
- B. INR
- C. GGT
- D. PSA

28. In which of the following cases do bilirubin, gammaglutamyl transferase, and alkaline phosphatase increase?

- A. Hepatocellular
- B. Cholestatic
- C. Cirrhosis
- D. Isolated hyperbilirubinemia

29. The study of different aspects of aging is known as

- A. Gerontology
- B. Gynaecology
- C. Odontology
- D. Chronology

30. Which of the following tissues or cells needs more time to completely replace or regenerate?

- A. Epidermis
- B. Leukocyte
- C. Adipocyte
- D. Cardiomyocyte

31. The sentence "The Brighter the Candle, the Quicker It Burns" is related to which of the theories of aging?

- A. Wear & Tear Theories
- B. Metabolic Theories
- C. Mitochondrial Theory
- D. Free Radical Theory

32. All of the following related to kidney function in old age are correct except:

- A. Renal function gradually deteriorates throughout life.
- B. Creatinine clearance falls.
- C. The ability of the kidneys to concentrate urine increases.
- D. The ability of the kidneys to excrete dilute urine declines.

33. Which anti-hyperlipidemic drug reduces lipolysis in adipose tissue and reduces VLDL production in the liver?

- A. Niacin
- B. Lovastatin
- C. Ezetimibe
- D. Cholestyramine

34. The mechanism of action of which of the following drugs is through increasing the activity of lipoprotein lipase?

- A. Clofibrate
- B. Cholestyramine
- C. Niacin
- D. Lovastatin

35. Which of the following is not an effect of Glucagon?

- A. Increase glycogenolysis
- B. Increase gluconeogenesis
- C. Increase ketogenesis
- D. Increase lipogenesis

36. All of the following are symptoms of type 2 diabetes except:

- A. Nephropathy
- B. Polyuria
- C. Ketoacidosis
- D. Obesity

37. All of the following are among diabetes diagnostic criteria except:

- A. Random glucose ≥ 100 mg/dL
- B. HbA1C $\geq 6.5\%$
- C. FBS ≥ 126 mg/dL
- D. 2h-plasma glucose ≥ 200 mg/dL

38. Which condition is characterized by high levels of cortisol?

- A. Addison's disease
- B. Cushing's syndrome
- C. Graves' disease
- D. Hashimoto's thyroiditis

39. Which hormone is primarily responsible for lowering blood glucose levels?

- A. Glucagon
- B. Epinephrine
- C. Insulin
- D. Cortisol

40. Selenium deficiency causes all of the following diseases, except ...?

- a. White muscle disease
- b. Menke's disease
- c. Goiter
- d. Keshan disease

41. A woman was admitted to hospital. ABG results are: pH 7.31; PCO₂ 33 mmHg; HCO₃⁻ 16 mmol/L; PO₂ 93 mmHg. Calculate the expected PCO₂ range.

- a. 27-31 mmHg
- b. 30-34 mmHg
- c. 33-37 mmHg
- d. 38-42 mmHg

42. In a 22-year-old man who has dehydration and pain in the stomach the electrolytes were measured and show the following results. Calculate the anion gap (AG) for this patient. Na⁺ 144; Cl⁻ 95; HCO₃⁻ 14

- a. 33
- b. 35
- c. 39
- d. 40

43. Which condition can cause an increase in blood pH?

- a. Hyperventilation
- b. Hypoventilation
- c. Hypernatremia
- d. Hyponatremia

44. A patient has low Na⁺ and his blood osmolarity was calculated to be normal. Which condition is the patient suffering from?

- a. Pseudohyponatremia
- b. True hyponatremia
- c. Isovolumic hyponatremia
- d. Hypernatremia

45. All of the following statements about biochemical changes during normal pregnancy is true, except?

- 1) The plasma globulin concentrations increase slightly.
- 2) Serum alkaline phosphatase activity is reduced.
- 3) The concentrations of ceruloplasmin and transferrin increase markedly.
- 4) The concentration of plasma albumin is decreased.

46. The marker pattern in Trisomy 18 during second trimester includes:

- 1) Low AFP- very low β -hCG- low uE3
- 2) Low PAPP-A- High β -hCG -high NT
- 3) High PAPP-A- very low β -hCG –low AFP
- 4) High AFP-Low β -hCG –high AFT

47. All of the following statements about Human Chorionic Gonadotropin is true, except?

- 1) A glycoprotein secreted by the syncytiotrophoblastic cells of the normal placenta.
- 2) The β subunit is common to LH, FSH and TSH.
- 3) The carbohydrate composition changes as pregnancy progresses.
- 4) When the hCG dimer is dissociated, hormone activity is lost.

48. Pregnancy Associated Plasma Protein A (PAPP-A) is a screening marker that produce by _____ and measurements are performed exclusively as part of _____ trimester aneuploidy screening.

- 1) fetus/ second
- 2) placenta/ second
- 3) fetus/ first
- 4) placenta/ first

49. Which of the following plasma proteins are negative acute-phase reactant?

- 1) Transferrin- Albumin-Ferritin
- 2) Hemopexin-Albumin-Procalcitonin
- 3) Transcortin- transferrin - Albumin
- 4) Albumin-Transferrin- Procalcitonin

50. In all of the following clinical conditions ceruloplasmin concentration decreases, except?

- 1) Blood loss
- 2) Pregnancy
- 3) Wilson's disease
- 4) Penicillamine therapy

51. Which of the following proteins bind free hemoglobin and delivers it to the reticuloendothelial cells?

- 1) Haptoglobin
- 2) Ferritin
- 3) Transferrin
- 4) transcortin

52. The main functions of albumin is:

- 1) Defense against bacteria and foreign substances
- 2) Acts as an antioxidant against free radicals
- 3) Transport iron and copper ions
- 4) Maintenance the oncotic pressure of plasma

نام و نام خانوادگی دانش آموز

شماره ۲۰

Attention!

- 1- If this answer-sheet is not yours, Please inform the responsible person
- 2- The answers should be marked with a soft and bold black pencil in the appropriate box.

How to Mark: Wrong ☐ ☐ ☐ ☐

Please don't write anything in

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